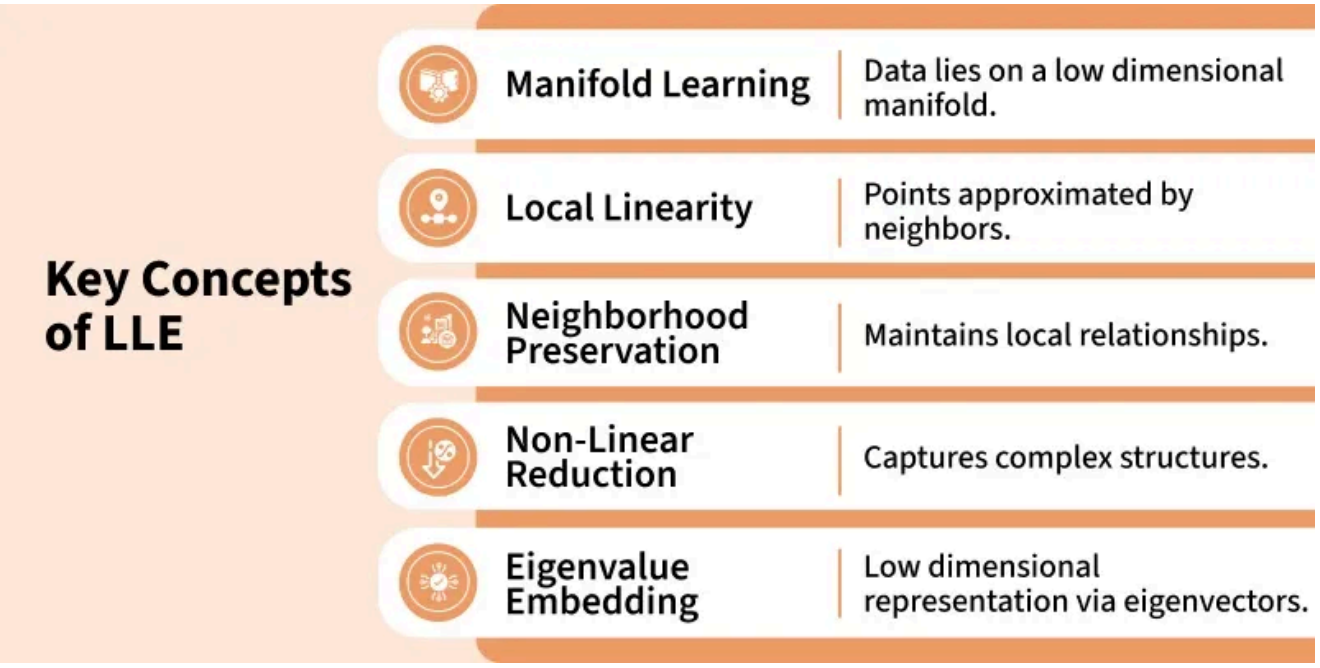
LLE = Locally Linear Embedding

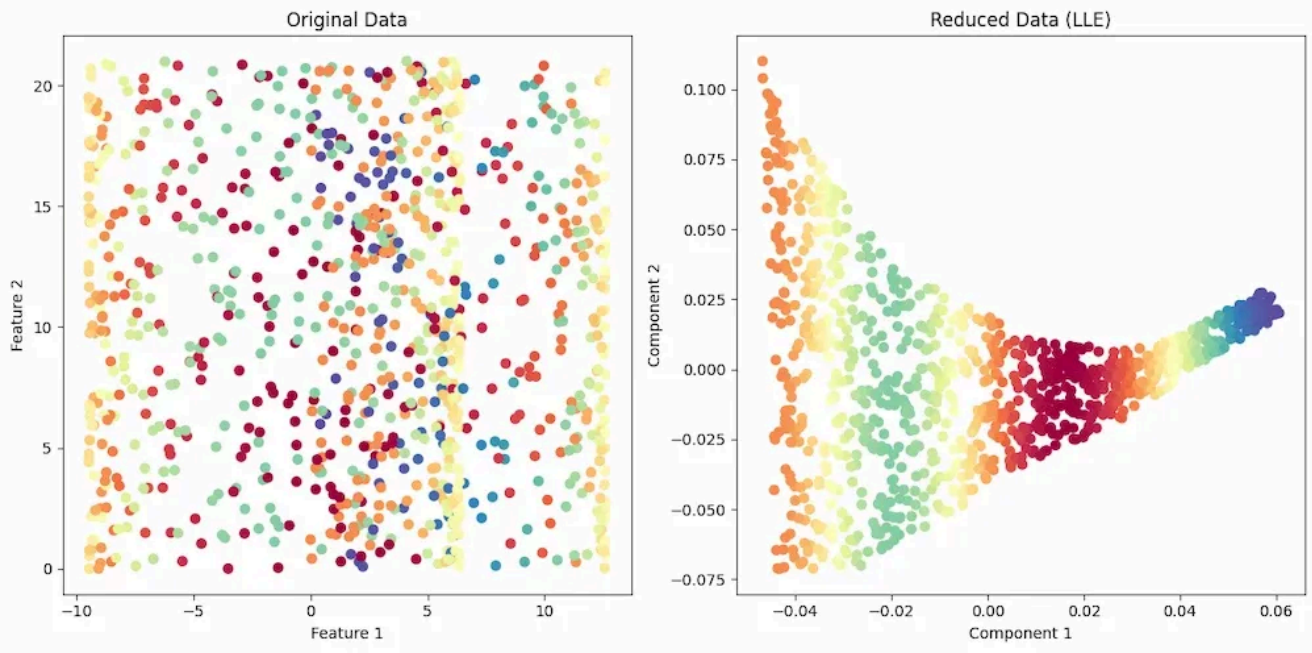
It is another very powerful **nonlinear dimensionality reduction** (NLDR) technique to uncover meaningful structure in high-dimensional data.
It is a manifold learning technique that does not rely on projections.

first, it measure how each training instance linearly relates to its closest neighbors, and then looking for a low-dimensional representation 2d or 3d of the training set where these local relationship are best preserved.




```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_swiss_roll
from sklearn.manifold import LocallyLinearEmbedding

n_samples = 1000
n_neighbors = 10
X, _ = make_swiss_roll(n_samples=n_samples)

lle = LocallyLinearEmbedding(n_neighbors=n_neighbors, n_components=2)
X_reduced = lle.fit_transform(X)

plt.figure(figsize=(12, 6))

plt.subplot(121)
plt.scatter(X[:, 0], X[:, 1], c=X[:, 2], cmap=plt.cm.Spectral)
plt.title("Original Data")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")

plt.subplot(122)
```

```
plt.scatter(X_reduced[:, 0], X_reduced[:, 1], c=X[:, 2], cmap=plt.cm.Spectral)
plt.title("Reduced Data (LLE)")
plt.xlabel("Component 1")
plt.ylabel("Component 2")

plt.tight_layout()
plt.show()
```



Application:

- Image processing
- Speech and Audio Analysis
- Data Visualization
- Manifold Learning in Biology
- Anomaly Detection

Pros:

- Preservation of local structures
- Handling non linearity
- Dimensionality reduction
- Good For visualization = project the data into 2d or 3d without losing much feature.
- unsupervised learning = does not require label so it work well for exploratory data analysis in various domain like images, speech or genetic

  Cons:
- Sensitive to parameters
- No out-of-sample mapping
- senstive to noise and outlier
- Loss of Global structure

- curse of dimensionality = In very high dimensional spaces, more neighbors are needed to capture local structures which increases computational costs.